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Under What Condition Can the Behaviour of One Network be Induced into Organisational Changes of another Suitably Sensitive Network?

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Abstract

This study investigates the role of inductive bias in the neural response of Multilayer Perceptron (MLP) model when it is trained using another isolated Perceptron classifier. The main research problem statement is: When one MLP classifier is used to train another MLP classifier, How the neural response of the last hidden layer in one neural network is affected by altering the inductive bias for the model that is used to train it? The studied inductive bias in this paper is the $L2$ Regularisation parameter λ. Initially, three main different cases are considered: (1) both models have an equally strong inductive bias, (2) both models have an equally weak inductive bias, and (3) both models have no λ inductive bias. The main hypothesis is that one neural network can cause relevant changes in the neural response of another neural network if they have an equally strong inductive bias. Results show that when two models are trained via Knowledge Distillation approach on the same task using the same ANN architecture, their neural response always reports an extent of similarity in the last hidden layers. However, this extent of similarity appears greater when the teacher model has a strong inductive bias and the student model has a weak one. In conclusion, models that are trained with an equally strong inductive bias exhibit a stronger connection than those which are trained with a weak one.